P3: cross-trace covariate test of the split-prime statistic

research/split_prime_ecdlp_20261107/followups/p3_cross_trace · run-20261107T2220Z · 2026-11-08

Outcome (frozen rule 2): null at the tested sizes [pcc:unverified]. Across 2,000 random prime-order curves at each of p_{24} = 16,777,213 and p_{28} = 268,435,399, neither exact $m = 2$ decomposition coverage nor the measured rho cost depends on z_{691} (split primes $\ell \le 691$), the partial $L(1, \chi)$ ratio, or the ψ_3/ψ_5 root counts: none of 14 tests is Bonferroni-significant (α = 0.003125), positive controls informative (planted +2 % recovered, power $\ge$ 0.995). The split-prime seed is closed for the coordinate track at these sizes. The A/A control exposed a defect in this study's rho walk, which limits the rho half. Card `CARD-p3-cross-trace-20261107-2`: executed as specified (director).

1. Method

Protocol frozen 535d9da826 with Addendum A (merged in #3657): K = 2,000 curves per prime, T = 256 rho targets per run, runs A and B with disjoint seeds, exact $m = 2$ coverage of the first N prime-abscissa points (N = 502 and 2,007), OLS of each outcome on each standardized covariate with $\log n$ as nuisance; ψ_5 dropped at p_{28} ($p_{28} \equiv -1 \bmod 5$ makes it identically zero for prime order). Instrument `p3census` (Rust). Card 1 was killed by the executor shell's 30-minute background limit after its p_{24} steps; card 2 ran detached. Replay check: card 1's p_{24} curves, covariates, rho A, rho B and analysis files are byte-identical to card 2's (yield identical apart from wall time).

2. Results

prime	outcome	z_{691}	logL1	ψ_3	ψ_5
p_{24}	rho/√n	+0.17 [−0.37, +0.71]	+0.37 [−0.17, +0.91]	−0.07 [−0.62, +0.47]	+0.70 [+0.16, +1.25]
p_{24}	coverage	−0.002 [−0.003, 0.000]	0.000 [±0.002]	−0.001 [±0.002]	0.000 [±0.002]
p_{28}	rho/√n	+0.26 [−0.45, +0.97]	+0.43 [−0.29, +1.15]	+0.61 [−0.11, +1.34]	not tested
p_{28}	coverage	0.000	0.000	−0.001 [−0.001, 0.000]	not tested

Effects per covariate SD, % of the outcome mean, 95 % CI; none Bonferroni-significant. Three rho intervals reach past +1 %, so the rho half cannot exclude a +1 % effect everywhere; no covariate is significant and $\ge$ 1 % at both primes.

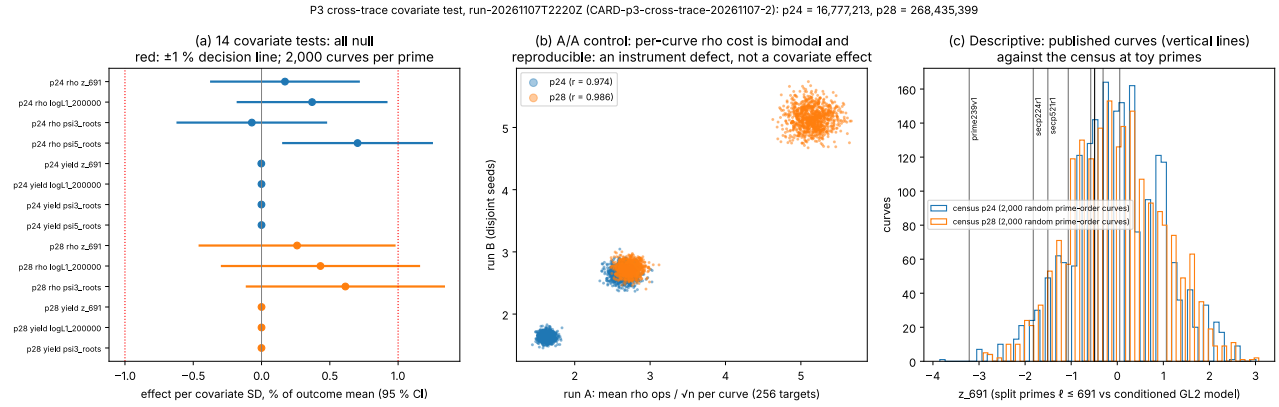


Figure 1: (a) the 14 tests with the ±1 % decision line; (b) A/A: per-curve rho cost in runs A and B; (c) census z_{691} with the published curves. Data from run-20261107T2220Z.

3. Instrument finding: the rho walk is defective

Runs A and B correlate at r = 0.974 (p_{24}) and 0.986 (p_{28}); between-curve variance is 41 and 72 times the within-curve expectation. Per-curve cost is bimodal, about 1.6 or 2.6 · √n at p_{24} and 2.7 or 5.1 · √n at p_{28}, while a sound negation-map walk costs about 0.9 · √n, flat in n. Outcome Y1 therefore measures this walk, not rho in general. The jump table is seeded per target; the clusters are not separated by trace parity, $t \bmod 4$, $n \bmod 3$, $n \bmod 4$ or $a \bmod 2$. Suspected ingredient: 32 small-coefficient jumps under the negation map. Problem record `problems/p3census-rho-walk-bimodal.md`. The coverage outcome is exact and rho-free.

4. Descriptive (decides nothing)

Percentile of each published curve's z_{691} in the census, p_{24} / p_{28}: prime239v1 0.05 % / 0.00 % (below all 2,000 at p_{28}), secp224r1 4.00 % / 2.75 %, secp521r1 5.50 % / 5.45 %, secp384r1 13.95 % / 14.50 %, others 27 % to 54 %.

5. Next steps

Re-run Y1 with the lab's `rho_negation` or a corrected walk under the same frozen covariates and seeds; diagnose the bimodality on 50 curves at $p \leq 2^{20}$.

6. Round 2: rho rerun with the corrected walk

Card `CARD-p3-cross-trace-20261108-r2` (executed as specified). Root cause of the round-1 defect: distinguished-point bits set per curve from $\lfloor \log_2 n/4 \rfloor$ at primes just below a power of two, plus long fruitless cycles from small jump coefficients. Walk v2 passed the frozen gate. **Outcome (PROTOCOL-R2 rule 3): null, instrument informative** [pcc:unverified]: A/A r = 0.027 / 0.009 (round 1: 0.974 / 0.986), variance ratios 0.95 to 1.02, every per-curve mean in 0.850 to 1.091 · √n; all 7 rho tests null, every 95 % interval within ±0.19 % of zero (largest −0.087 % per SD, p_{24} z_{691}, p = 0.083), minimum detectable effect 0.19 %. With round 1′s coverage null, the seed is closed on both outcomes at 2^{24} and 2^{28}.